# The MCDM Algorithm

## A Scientifically-Grounded Multi-Criteria Decision Making Engine for a Crop Rotation Decision-Aid Hub

NASA Space Apps Challenge 2026 — Crop Rotation Decision-Support Challenge

Implementation: Raspberry Pi 4 hub · NASA POWER + ISRIC SoilGrids · FAO EcoCrop · FAO-56 · AquaCrop · AHP/SAW/TOPSIS · dynamic-programming rotation planner

Technical Report — what MCDM is, the exact mathematics, and how it is implemented in this project

## 1. Executive summary

Choosing a crop when several conflicting criteria matter is a **Multi-Criteria Decision Making (MCDM)** problem, and this project implements it inside a **Decision Support System (DSS)** for crop selection, following the paper the team found (Gunawan, Sitopu & Sechan, 2024). That paper's mathematics is the **weighted-sum (Simple Additive Weighting, SAW)** model: each crop is scored on several criteria; each criterion has an importance weight; the crop with the highest weighted total is recommended.

This report documents (a) what MCDM/SAW is mathematically, (b) how the paper's formulation is generalised into a rigorous, fully-specified decision pipeline — AHP weights → SAW ranking → TOPSIS verification → dynamic-programming multi-season rotation planning, and (c) how every criterion score is computed from established agronomy: the FAO EcoCrop suitability model (exact OpenCLIM formulation), FAO-56 crop water requirements, ISRIC SoilGrids soil data, and the FAO AquaCrop yield response-to-water model. All of it is driven by NASA POWER satellite data cached on a Raspberry Pi 4 for offline rural use.

The implemented engine is validated by a test suite that **reproduces the source paper's numerical example exactly** (crops A, B, C → suitability 0.76, 0.69, 0.63; winner A), alongside unit tests of every physical equation and a brute-force optimality check of the rotation planner.

## 2. MCDM inside a DSS — the method this hub implements

MCDM (also MCDA — multi-criteria decision analysis) is the discipline of making choices when several conflicting criteria matter simultaneously. The Gunawan et al. (2024) paper formalises crop selection as an optimisation:

$$\text{maximize} \quad Z = \Sigma_i \, \Sigma_j \, W_j \cdot C_{ij} \cdot X_{ij}$$

where **i** indexes crops, **j** indexes criteria, $\mathbf{W_j}$ is the importance weight of criterion j, $\mathbf{C_{ij}}$ is the suitability score of crop i on criterion j, and $\mathbf{X_{ij}} \in \{0,1\}$ selects which crop–criterion pairs count. Because exactly one crop is selected, every row uses a single X, and the model collapses to the classic **Simple Additive Weighting (SAW)** score per crop:

$$Z_i = \Sigma_j W_j \cdot C_{ij} \quad (\text{choose } \operatorname{argmax}_i Z_i)$$

The paper's worked example (weights: climate 0.3, soil fertility 0.2, market demand 0.4, water 0.1) gives $Z_A = 0.76$, $Z_B = 0.69$, $Z_C = 0.63$ → Crop A wins. Our test suite reproduces these numbers to machine precision, proving the implemented core is faithful to the paper.

What the paper leaves unspecified — and what this project therefore specifies rigorously — is: **(1)** how the weights $W_j$ are derived from a farmer's priorities (answer: Analytic Hierarchy Process, Saaty 1980, with a consistency check), **(2)** how each $C_{ij}$ is computed from real geophysical data (answer: EcoCrop, FAO-56, SoilGrids, AquaCrop), **(3)** whether the ranking is robust (answer: cross-check with TOPSIS), and **(4)** how to plan not just the next crop but a whole multi-season rotation (answer: dynamic programming over a documented transition-score matrix).

# 3. The implemented decision pipeline

Step 0 — **Data**: NASA POWER daily point series (T2M, T2M_MAX, T2M_MIN, PRECTOTCORR, WS2M, RH2M, ALLSKY_SFC_SW_DWN) and ISRIC SoilGrids soil properties (pH, sand, clay, soil organic carbon over 0–60 cm), cached locally in SQLite and refreshed every 24 hours; every feature keeps working from cache when the hub is offline.

Step 1 — **Candidate crops**: the FAO EcoCrop database (2 568 crops with temperature, precipitation, soil and cycle parameters) is bundled on the hub; the farmer shortlists crops or uses the curated default set of major food crops.

Step 2 — **Criterion scores** $c_{ij} \in [0,1]$ per crop (§4).

Step 3 — **Weights** w from AHP on the farmer's priority sliders (§5.1).

Step 4 — **Ranking**: SAW (the paper's objective), verified by TOPSIS (§5.2–5.3); both rankings are shown to the farmer.

Step 5 — **Rotation plan**: exact dynamic programming over seasons with agronomic transition scores (§6).

# 4. The criterion scores $c_{ij}$

## 4.1 Climate suitability — FAO EcoCrop (exact OpenCLIM formulation)

EcoCrop scores how well a crop's thermal and rainfall requirements are met. Following the OpenCLIM/ecocrop implementation (Bib. [4]), for each candidate growing window of GTIME days (GTIME stepped in 10-day increments between the crop's cycle limits GMIN..GMAX), each day t contributes an intermediate temperature score:

$$D(T) = (T - TMIN)/(TOPMN - TMIN) \text{ when } TMIN < T < TOPMN$$
$$D(T) = 1 \text{ when } TOPMN \le T \le TOPMX$$
$$D(T) = (TMAX - T)/(TMAX - TOPMX) \text{ when } TOPMX < T < TMAX$$
$$D(T) = 0 \text{ otherwise}$$

where T is the day's mean temperature and TMIN, TMAX, TOPMN, TOPMX are the crop's EcoCrop parameters. The window accumulates suitable days:

$$N_D = \Sigma_{\text{window}} D(T)$$

and the temperature suitability of that window is:

$$S_T = 100 \cdot [1 - (GTIME - GMIN)/(GMAX - GMIN)] \text{ if } N_D \ge GMIN, \text{ else } 0$$

— i.e. the faster the crop can accumulate its minimum number of suitable days, the higher the score. Two penalties are then applied: any night below the killing temperature KTMP zeroes S_T (frost kill), and each day with maximum temperature above TMAX subtracts one point (heat stress). For perennial crops the window-mean temperature is scored instead (documented in [4]).

Precipitation suitability uses the window rainfall total PTOT:

$$S_P = 100 \cdot (PTOT - RMIN)/(½(ROPMN+ROPMX) - RMIN) \quad \text{below optimum}$$
$$S_P = 100 \cdot (RMAX - PTOT)/(RMAX - ½(ROPMN+ROPMX)) \quad \text{above optimum}$$
$$S_P = 0 \quad \text{outside [RMIN, RMAX]}$$

The combined EcoCrop score is **S = min(S_T, S_P)** — "the lowest score is likely to be the limiting factor". The hub scans all candidate sowing windows over the cached year of NASA POWER data and reports the best window as the recommended sowing date (the OpenCLIM aggregation takes the maximum over window lengths, and a 95th-percentile when aggregating climatologies; for a single field decision the maximum is the appropriate analogue).

**Rainfed vs irrigated.** Under rainfed conditions the hub uses the exact OpenCLIM min(S_T, S_P): rainfall adequacy is part of climate fit. When the farmer declares an irrigation supply (mm available), the climate criterion becomes the best temperature score S_T alone — irrigation removes the rainfall limitation — and water adequacy is instead scored by the FAO-56 criterion below. This distinction is a design decision of the hub, documented because it changes interpretation in arid regions.

## 4.2 Soil suitability — SoilGrids vs EcoCrop soil ranges

SoilGrids 2.0 (ISRIC) provides soil pH (phh2o), sand, clay and organic carbon at standard depths; the hub averages 0–60 cm (root zone). Each property is scored with a trapezoid membership function against the crop's EcoCrop soil requirement ranges — 1 inside the optimal range (PHOPMN..PHOPMX for pH), linearly decreasing to 0 at the absolute limits (PHMIN..PHMAX); texture classes (heavy/medium/light/organic, derived from the USDA sand–clay triangle) are scored 1 if in the crop's optimal texture list, 0.5 if in its tolerated list, else 0; rooting depth is scored against the crop's DEP class. Scores are aggregated by a weighted mean (weights: pH 0.4, texture 0.3, depth 0.2, salinity 0.1 when measured EC is available):

$$S_{soil} = \Sigma_j \, w_j \, S_j, \quad \Sigma \, w_j = 1$$

This aggregation is a design choice of the system (as the team's research notes correctly state, it is not a universal law); the per-property functions themselves follow the EcoCrop requirement ranges, and the weights are exposed and documented for calibration.

## 4.3 Water adequacy — FAO-56

Reference evapotranspiration is computed daily with the FAO-56 Penman–Monteith equation from NASA POWER inputs (Allen et al. 1998, Bib. [5]):

$$ET_0 = [0.408 \cdot \Delta \cdot (R_n - G) + \gamma \cdot (900/(T+273)) \cdot u_2 \cdot (e_s - e_a)] \,/\, [\Delta + \gamma \cdot (1 + 0.34 \cdot u_2)]$$

with the net radiation $R_n$ built from the measured solar radiation $R_s$ ($R_{ns} = 0.77 \cdot R_s$; $R_{nl}$ via the Stefan–Boltzmann cloudiness correction), $e_s$ from daily Tmax/Tmin, $e_a$ from RH, and $u_2$ the 2 m wind speed. A Hargreaves–Samani fallback ($ET_0 = 0.0023 \cdot R_a \cdot (T_{max} - T_{min})^{0.5} \cdot (T_{mean} + 17.8)$) covers incomplete records (documented overestimation bias in arid, high-radiation conditions).

Crop evapotranspiration follows the FAO-56 stage curve ($K_c$ initial/mid/late trapezoid, FAO-56 Table 12 values bundled for ~30 major crops):

$$ET_c = K_c(\text{stage}) \cdot ET_0$$

Effective rainfall uses the FAO "80% rule" (Dastane 1978): $P_e = 0.8 \cdot P$ for $P < 75$ mm, else $0.6 \cdot P + 15$. The seasonal net irrigation requirement is:

$$NIR = \max(0, \Sigma \, ET_c - \Sigma \, P_e)$$

In irrigated mode the water criterion is the supply fraction min(1, available/NIR); in rainfed mode it equals the EcoCrop precipitation suitability S_P (§4.1), the honest measure of rainfed adequacy. In all cases NIR is displayed as the advisory irrigation volume.

## 4.4 Yield response to water — simplified AquaCrop

AquaCrop's growth engine is water-driven (Steduto et al. 2009, Bib. [6]): daily transpiration translates into biomass through the normalised biomass water productivity WP*, and yield is the harvestable fraction:

$$B = WP^* \cdot \Sigma\,(Tr_i / ET_{0,i}) \quad Y = HI \cdot B$$

WP* is conservative per functional group (C4 cereals ≈ 33.7 g $m^{-2}$ — the AquaCrop default for maize; C3 cereals ≈ 15–20; legumes ≈ 15; the hub's bundled table gives values per group). In the simplified screening implementation, without a daily canopy–water-stress simulation, transpiration is estimated from the Kc curve as $Tr_i \approx ET_{c,i} \cdot (1 - f_e)$ with $f_e = 0.30$ (soil-evaporation share of Kc, documented). The resulting yield (t/ha) is used as a relative criterion — normalised against the best-yielding crop in the candidate set — and is explicitly labelled a screening estimate, not a calibrated AquaCrop simulation.

### 4.5 Rotation compatibility — pre-crop effects

Agronomic literature demonstrates that the previous crop changes the performance of the next one (pre-crop values; rotation matrices — Fenz et al. 2023 found data-driven matrices agree with literature recommendations in ~52% of pairs and within one degree in 97%; functional-group diversity raises cereal yields — Smith et al. 2023). The hub encodes a transition score T(prev → cur) ∈ [−1, 1]:

| Transition | Score | Rationale |
|---|---|---|
| same botanical family | −1 (forbidden) | pest/disease carry-over; nutrient imbalance (family-repeat constraints a |
| same crop | −0.5 | monoculture penalty |
| legume (Fabaceae) → non-legume | +0.3 | residual N credit of legumes (~30–80 kg N/ha) |
| deep-rooted → other | +0.1 | subsoil structure / nutrient capture |

The rotation criterion score used inside the MCDM matrix is $c_{rot} = ½(1 + T) \in [0,1]$.

### 4.6 Market demand

The farmer rates expected market demand per crop on a 0–1 slider (default 0.5, neutral). This keeps the economic criterion of the source paper (market demand, weight 0.4 in its example) in the farmer's hands, since reliable price data are not yet integrated.

## 5. Weights (AHP), ranking (SAW) and verification (TOPSIS)

### 5.1 Analytic Hierarchy Process — deriving the weights

Farmers should not have to type decimal weights — or understand criteria at all. The farmer answers five plain-language priority questions on a 1–5 scale ("not important" to "very important"): *Fits my land and weather*, *Big harvest*, *Saving water*, *Healthy soil next seasons*, and *Sells for a good price*. These map onto the six science criteria ("fits my land and weather" deliberately drives BOTH the climate and the soil criterion with the same weight — the farmer thinks of it as one question), and the 1–5 scale is stretched to the 1–9 AHP scale ($p \to 2p - 1$, i.e. 1, 3, 5, 7, 9) so "very important vs not important" keeps its intended strength in the pairwise matrix. The hub builds the pairwise comparison matrix A with entries $a_{ij} = \text{snap}(p_i/p_j)$ to the Saaty scale {1/9, …, 1, …, 9}, and computes the **principal eigenvector**:

$$A \cdot w = \lambda_{max} \cdot w\,, \quad \Sigma\, w_j = 1$$

Consistency is measured with Saaty's index and random index RI:

$$CI = (\lambda_{max} - n)/(n - 1)\,, \quad CR = CI/RI(n)$$

with RI: n=3 → 0.58, 4 → 0.90, 5 → 1.12, 6 → 1.24, 7 → 1.32, 8 → 1.41, 9 → 1.45, 10 → 1.49. A CR ≤ 0.10 is accepted (Saaty's rule); the UI warns otherwise. The hub's typical CR ≈ 0.01 — well within tolerance.

### 5.2 Simple Additive Weighting — the MCDM objective

$$Z_i = \Sigma_j\, w_j \cdot c_{ij} \;\rightarrow\; i^* = \text{argmax}_i\, Z_i$$

This is exactly the paper's objective $Z = \Sigma\Sigma\ W_j \cdot C_{ij} \cdot X_{ij}$ under its single-crop constraint (the binary selection variables add nothing when one alternative is chosen; the argmax form is the mathematically equivalent, standard SAW statement). Because all criteria are normalised to [0,1] and $\Sigma w = 1$, $Z \in [0,1]$ is interpretable as an overall suitability.

### 5.3 TOPSIS — independent verification

To guard against weighting artefacts, the hub also ranks crops with TOPSIS (Hwang & Yoon 1981): vector-normalise the decision matrix ($v_{ij} = c_{ij}/\sqrt{\Sigma_i c_{ij}^2}$), weight by $w_j$, take the ideal $a^+$ and anti-ideal $a^-$ columns, compute Euclidean distances $d^+$, $d^-$ per crop and the closeness coefficient:

$$C_i = d^-_i \,/\, (d^+_i + d^-_i)$$

SAW and TOPSIS agree on the winner in practice; when they disagree the UI flags it (a genuinely borderline decision, worth the farmer's attention). All hub criteria are benefit-type (higher is better); any future cost criterion is pre-inverted.

## 6. Multi-season rotation planning — exact dynamic programming

Choosing the next crop is a myopic decision; a rotation must also be good over several seasons. Let $Z_i$ be the SAW score of crop i and T(a→b) the transition score (§4.5). The hub chooses the sequence $s_1 \ldots s_K$ over K seasons (default K = 3) maximising:

$$J = \Sigma_k\, Z_{s_k} + \lambda \cdot \Sigma_{k \geq 2}\, T(s_{k-1} \rightarrow s_k)\ , \quad \lambda = 0.5$$

subject to T = −1 transitions (same botanical family) being forbidden. This is solved **exactly** by dynamic programming over a layered graph:

$$dp[1][i] = Z_i$$
$$dp[k][b] = \max_{a:\ T(a,b) > -1} \{\, dp[k-1][a] + \lambda \cdot T(a,b) \,\} + Z_b$$

with back-pointers recovering the optimal sequence; complexity $O(K \cdot n^2)$. A brute-force enumeration test confirms the DP reaches the true optimum (equivalent to a longest-path formulation; unlike greedy first-pick it never falls into locally-good, globally-poor sequences). The hub also reports a screening-level nitrogen balance over the plan: legumes credit ≈ +60 kg N/ha, non-legumes debit ≈ −45 kg N/ha per season.

## 7. The hub system

| Component | Implementation | Notes |
|---|---|---|
| Hardware target | Raspberry Pi 4 | backend: Python stdlib + numpy (+ requests for data fetches) |
| Web interface | Farmer-friendly single-page app, port 8080 | plain language throughout; no math exposed; four numbered s |
| Deployment | systemd service + venv (INSTALLED and verified) | sudo ./install.sh creates the venv (numpy), installs systemd/lcr |
| Climate data | NASA POWER daily point API (7 variables) | cached in SQLite; 24-hour refresh (background thread); one rec |
| Map imagery | NASA GIBS WMTS (MODIS true colour, daily imagery, ~1 km) | SMAP soil moisture, MODIS LST, VIIRS/ADV, AMER, disk-cached |
| Soil data | ISRIC SoilGrids 2.0 REST (pH, sand, clay, SOC, 3060 cm) | handles NULL cells (rivers, dense cities) by sa |
| Crop parameters | FAO EcoCrop database, 2 568 crops | bundled CSV (FAO EcoCrop / OpenCLIM, OGL v3) |
| Decision engine | AHP → SAW → TOPSIS + DP rotation | fully deterministic, reproducible; when SAW and TOPSIS disagre |
| Offline behaviour | all features run from cache | designed for farmers without internet |

API surface: GET /api/health, GET /api/crops?q=…, POST /api/recommend (lat, lon, shortlist, farmer priorities {land, harvest, water, soilnext, market} on 1–5 — raw 1–9 criterion keys also accepted, previous crop, water available in mm (null = rainfed), market scores 0.2–1.0; the response echoes

waterMode/waterAvailableMM and includes closeCall when the two methods disagree), POST /api/refresh (fair-use guarded: refetches only when the cache is stale).

### 7.1 Validation

| Test | Expected | Result |
|---|---|---|
| Paper numerical example (SAW) | Z_A=0.76, Z_B=0.69, Z_C=0.63, winner A | PASS (machine precision) |
| EcoCrop daily/window equations | piecewise scores, frost zeroing | PASS |
| FAO-56 $ET_0$ / Kc / NIR | plausible $ET_0$, Kc trapezoid, Pe rule | PASS |
| AquaCrop B = WP*·Σ(Tr/$ET_0$), Y = HI·B | exact algebra | PASS |
| AHP consistency diagnostics | CR ≤ 0.1 detection | PASS |
| Rotation DP vs brute force | identical optimum | PASS |
| End-to-end live run (Egypt, 29.2 N, 31.0 E) | 3 crops for Nov sowing; AHP CR 0.0092; SAW=TOPSIS winner | PASS |

35/35 tests pass (mcdm-hub/tests/test_mcdm.py). The end-to-end run used live NASA POWER data (366 days cached, $ET_0$ mean 6.79 mm/day) and live SoilGrids data (pH 8.0, clay 24.8 %, sand 47.5 %, SOC 4.8 %).

## 8. Known limitations (documented, not hidden)

1. **Screening-level yield**: the AquaCrop implementation is the conceptual equation without the full daily soil-water–canopy simulation; treat yields as relative indicators. Calibrate with local data before claiming quantitative accuracy (the team's own Resources.md demands this).

2. **Simplified seasonal water balance**: NIR uses effective rainfall, not a daily soil-water-bucket model; adequate for ranking, not for irrigation scheduling.

3. **SoilGrids is modelled data** (~1 km): the hub accepts measured pH (and could accept EC) to override.

4. **Weights and transition scores** are design choices documented for calibration, not universal laws — consistent with the MCDM literature's advice to elicit weights from stakeholders.

5. **EcoCrop rainfall suitability is rainfed-oriented**: hence the explicit rainfed/irrigated mode (§4.1).

6. **Market criterion** is farmer-entered until a price data source is integrated.

7. **24-hour cache** is a fairness/simplicity trade-off; POWER updates daily, so no information is lost.

## 9. References


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Appendix A — repository map: mcdm-hub/run.py (entry) · backend/engine/ (mcdm, ecocrop, fao56, soil, aquacrop, rotation) · backend/data/ (power, soilgrids, crops + EcoCrop_DB.csv) · backend/storage.py (SQLite cache) · backend/app.py (pipeline) · backend/server.py (HTTP) · frontend/ (UI) · tests/test_mcdm.py (35 tests) · agent.md (full research & work log).